

Bank Loan Portfolio Analysis

Project report by Subachan Subedi



Project purpose and scope

This project uses Microsoft Excel to explore 38,576 consumer loans issued during 2021. It helps a reader understand lending activity, repayment status, and the types of borrowers and loans in the portfolio. The main result is that 86.18% of records are Current or Fully Paid, while 13.82% are Charged Off. These categories describe the recorded loan status; they do not predict future defaults.

A bank needs more than a list of individual accounts to review its portfolio. It needs a clear view of loan counts, funding, recorded payments, and areas where lending is concentrated. This workbook brings those views together in a Summary dashboard and an Overview dashboard. Grade and purpose slicers allow readers to explore the connected analysis.

Domain knowledge

A bank loan provides money that a borrower agrees to repay over time, usually with interest. Loans can support personal needs or business plans. Borrowers need to understand the repayment schedule, interest, fees, and other conditions. For a bank, the central question is whether the borrower can repay the loan as agreed.

How banks collect loan data

Banks gather personal and financial details from loan applications, credit reports, and their own transaction records. Online banking portals add application and payment activity. External verification services may provide income or employment information. Together, these sources help a bank understand the applicant and keep track of the loan after approval.

Why banks analyse loan data

Loan analysis supports lending decisions and portfolio monitoring. It helps banks compare borrower groups, detect unusual activity, review repayment problems, and prepare required reports. Banks also use broader financial and customer data to study profitability, improve products, plan lending activity, and identify refinancing or customer-retention opportunities. This project focuses on portfolio performance and borrower patterns; it does not build a fraud detector or a default-prediction model.

How a loan moves from application to repayment

- 1. Application and document review.** The customer applies for a loan. The bank reviews the application and supporting identity, income, and financial documents.
- 2. Identity and credit checks.** The bank verifies who the applicant is and reviews credit history, existing borrowing, and previous payment problems.
- 3. Affordability checks.** Income, employment, existing debt payments, and the debt-to-income ratio help the bank assess repayment capacity. A lower DTI generally leaves more income available for other expenses and payments.
- 4. Collateral and overall risk review.** For a secured loan, the bank assesses the proposed collateral. It also considers the loan purpose, income stability, and the applicant's wider financial position.
- 5. Decision and agreement.** The bank approves or declines the application. If approved, it sets the amount, rate, term, fees, and repayment schedule in a loan agreement.
- 6. Disbursement and repayment.** After the required agreements are completed, the bank releases the funds. The borrower makes the scheduled principal and interest payments.
- 7. Ongoing monitoring.** The bank tracks payments and changes in loan status, follows up on missed payments, and reviews emerging repayment problems.

Data used in the project

The supplied loan table contains 25 fields, including a derived Good Vs Bad Loan field. It includes loan and member identifiers, state, employment length, home ownership, income, purpose, credit grade, loan term, interest rate, debt-to-income ratio, loan amount, payment amount, and important dates. The original repository describes the source as a public Kaggle banking-loan dataset, but it does not provide the exact dataset listing. This report uses the attached workbook as its source.

The Data folder provides the records in two formats. The CSV contains the existing values and uses readable dates in YYYY-MM-DD format. The Excel data file keeps the source table and its loan-classification formula. Both files contain the 38,576 loan records and exclude the additional totals row below the original table.

Understanding the loan fields

The fields below describe the borrower, loan terms, and repayment history. They support grouping and comparison in this project. A single field, such as job title or home ownership, does not establish a borrower's credit risk by itself.

Field	Meaning and use
Loan ID and member ID	Identify the individual loan and its associated member or borrower record.
Address state	Shows the borrower's location and supports geographic comparisons of loan volume and performance.
Application type	Describes the application category recorded in the dataset.
Employment length and title	Describe the borrower's employment history and job. Banks can use these details alongside income verification.
Grade and sub grade	Record the lender's risk categories. Sub grades provide a more detailed classification within each grade.
Home ownership	Describes housing status, such as renting, a mortgage, or ownership. It does not by itself confirm that a loan is secured.
Issue date	Records when the loan was issued. It supports monthly trends, loan-age analysis, and grouping by issue period.
Last credit pull date	Records when the borrower's credit information was last accessed or updated in the dataset.
Last and next payment dates	Identify the latest recorded payment and the next scheduled payment date. They help track payment timing.
Loan status	Shows the recorded loan state, such as Current, Fully Paid, or Charged Off.
Good Vs Bad Loan	The workbook's derived category: Current or Fully Paid are Good Loans; Charged Off are Bad Loans.
Purpose and term	Describe why the loan was taken and its repayment length in months. These fields support loan-segment comparisons.
Verification status	Indicates the recorded verification status of the borrower's financial information.
Annual income	Records yearly income, which helps lenders assess repayment capacity alongside debt and other obligations.
Debt to income ratio	Compares monthly debt payments with monthly income. It gives an indication of the borrower's existing debt burden.
Instalment	The scheduled loan payment, generally covering principal and interest. It helps describe the ongoing payment commitment.
Interest rate	The stated annual interest rate used to price borrowing. Fees can also affect the total cost of the loan.
Loan amount	The loan principal recorded in loan_amount. Its total is used for funded amount in this report.
Total payment	Payments recorded in total_payment. Its total is used for amount received, rather than profit.
Total accounts	The account-count field recorded for the borrower; it is separate from the number of loans in this project.

How the workbook works

The workbook has a simple flow. The Bank Loan Data sheet holds the native Excel table called Table1. The Design Sheet groups and summarizes that table through PivotTables and formulas. The SUMMARY DASHBOARD and OVERVIEW DASHBOARD display the results through linked KPI shapes and charts. A linked shape reads a cell result and presents it as a large number or label.

Sheet	Purpose
Bank Loan Data	Individual records and the derived loan-status category
Design Sheet	Supporting PivotTables, formulas, and KPI calculations
SUMMARY DASHBOARD	Portfolio KPIs, Good and Bad Loan views, and status comparisons
OVERVIEW DASHBOARD	Monthly trends, state map, terms, purpose, and borrower profiles

A PivotTable groups records by a selected field and calculates summaries such as counts, sums, or averages. PivotCharts turn those summaries into visual comparisons. GETPIVOTDATA retrieves results from a PivotTable for reporting. Slicers provide buttons for filtering connected PivotTables and charts by grade or purpose. Clear the slicers before comparing a filtered view with the full-portfolio figures in this report.

Metric definitions

Loan applications count the individual records. Funded amount means the sum of loan_amount, and amount received means the sum of total_payment. Average interest rate and average debt-to-income ratio are simple averages across loans. Debt-to-income ratio describes the borrower's debt burden relative to income. These averages are not weighted by the size of each loan.

The workbook classifies Current and Fully Paid loans as Good Loans. Charged Off loans are Bad Loans. A Current loan is still active and may later change status, so the word Good should be read as a reporting category. The classification is useful for an overview, but it is not a complete credit-risk assessment.

Monthly measures and visual views

MTD means month to date. MoM means the percentage change from the previous month, usually calculated as the current-month value divided by the previous-month value, minus one. The supplied calculation sheet uses November and December 2021 for its comparison. It is a historical setup and does not automatically represent the current calendar month.

The Summary dashboard focuses on portfolio size and loan status. The Overview dashboard adds the monthly application trend, a native Excel state map, a term donut chart, category bar charts, and a home-ownership treemap. The existing DETAILS navigation button opens the data sheet; the workbook does not contain a separate Details dashboard.

Portfolio results

The table below uses the field names in the source loan table. The refreshed dashboard now follows the same funding and payment definitions. All figures here include every loan record, with no slicer filters applied.

Measure		Result	
Loan applications		38,576	
Funded amount from loan_amount		\$435,757,075	
Amount received from total_payment		\$473,070,933	
Average interest rate		12.05%	
Average debt-to-income ratio		13.33%	
Current or Fully Paid loans		33,243 86.18%	
Charged Off loans		5,333 13.82%	
Loan status	Records	Funded amount	Amount received
Fully Paid	32,145	\$351,358,350	\$411,586,256
Current	1,098	\$18,866,500	\$24,199,914
Charged Off	5,333	\$65,532,225	\$37,284,763

Lending patterns and borrower profiles

Monthly applications rise from 2,332 in January to 4,314 in December, an increase of about 85.0% between those two months. There is a small decline from January to February, followed by higher monthly counts through the rest of the year. This shows increasing application volume in the recorded issue-year data, rather than proof that credit quality improved.

Debt consolidation is the largest purpose category with 18,214 loans, or 47.22% of records. Credit card loans follow with 4,998 records. A total of 28,237 loans have a 36-month term, compared with 10,339 loans with a 60-month term. The shorter term represents 73.20% of the portfolio.

Renters account for 18,439 records and mortgage holders for 17,198. Together, these groups represent 92.38% of borrowers. The largest employment-length group is 10 or more years, with 8,870 records. California has the highest state count at 6,894, followed by New York at 3,701. These are volume patterns; separate charge-off comparisons are needed to assess each segment's risk.